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Nonlinear Trends in Modern Artificial Intelligence: A New Perspective

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Abstract. The paper faces the challenge to generalize existing trends and approaches in the field of artificial intelligence. The most perspective research directions are revealed through multidisciplinary analysis. It was shown that dynamic models with unstable dynamics are logical continuation of approaches sequentially becoming more complex. Conclusion is that in the nearest future nonlinear dynamics and chaos will become the most demanded apparatus to understand and model cognition processes.

Keywords: multidisciplinary analysis, distributed AI, bio-inspired algorithms, nonlinear dynamics, chaotic neural networks, self-organization

1 Introduction

This paper analyses trends of modern approaches (including methods partially) in artificial intelligence (AI) and change (transformation) of the multidisciplinary role and its level necessary in order to attain a qualitative leap in AI area. It is still a matter of future research to create true human intelligence although a lot of promising results in some local AI areas were published [5, 11, 19]. We aim to generalize up to date accumulated experience in intelligent systems development. To reach this aim it is necessary to stay above separated and fragmented multidisciplinary studies and to find the common approach which will let to aggregate them in the way to obtain synergy effect. We focus on the essence of AI paradigms and discuss on promising directions in developing of true human intelligence.

Two main approaches within AI field, namely Symbolic AI (associated with experts systems and knowledge bases, etc.) and Connectionist AI (associated with neural networks (NN) and implicit knowledge learning, etc.) no more compete but supplement each other [8, 14], e.g. in the form of neuro-fuzzy

methods when both learning processes and explicit knowledge statements are combined.

However, despite this mutual enrichment AI theory is still far from its ambitious goal that seemed to be so quick and easy to reach creation of genuine intelligent systems [1, 3, 4, 15] capable to learn, set goals and solve problems without external assistance, find new solutions that were not foreseen at the development stage.

Therefore it is natural to look on the brain, its texture and principles of functioning, because exactly brain makes it possible to solve intellectual tasks.

2 Brain Science and Limitations of Basic AI Approaches

Latest promising results in the field of genetic engineering together with wide accumulation of experimental data on brain dynamics obtained by means of modern techniques in noninvasive supervision over brain functioning [9, 10, 20] extended greatly the knowledge on cortex activities. High performance computing makes possible to model and in detail examine memory and behavioral processes that are accomplished by cells. A lot of research activities deal with estimation of chemical compound concentrations and electrical potential levels that indicate different states of neural systems [5, 9, 10]. It is worth mentioning that neural equations that describe neuron functioning were deduced in the late 1940's. It would seem that having knowledge on the structure and properties of basic construction element it is possible to extend this knowledge to create similar artificial intelligent system. However time passed but things are nearly right where they started. Only recently there were obtained results on part of rat brain modeling and on cultivation of biological neurons and networks on substrates (biological chips). It's time to integrate all practical AI knowledge and knowledge of relative scientific fields by usage of system analysis.

But wide knowledge about system micro-level functioning seldom leads to understanding a system as a whole entity, because it is necessary to take into account synergetic effects. This way helps to get and understand total system picture and interactions inside the system [4, 6, 15, 18].

So the dilemma sounds like this: at what extent we should fix our efforts on biological processes imitation and at the same time what level of abstraction from origin prototype is bearable?

There is a huge amount of isolated single-purpose AI models and methods [11, 19] that are effective in narrowly defined problems. But it is hard to

consider them separately as basic ideas for general theory. Thus there are a lot of methods, great theoretical contributions and empirical materials deeply worked through but general picture happens to be rather messy.

It is obvious that idea to combine all existing artificial intelligence models into the unified super one wouldn't give a positive effect. And here arises more particular research question — what trend is more perspective from the point of development general AI theory.

One of the main issues that challenge Symbolic AI researchers consists in impossibility to formalize the representation of all variants of situations a system can meet during functioning [14, 15]. After making some assumptions a set of problems of a certain class (dependent on the limitations) are successfully solved but universality appears to be unattainable - the system is intelligent at the extent that is predetermined by input data comprehensiveness and predesigned scenarios. But main task is to get generation of new knowledge inside the system even if input data are not complete or contradictory.

Connectionist AI researches have faced the same restriction on representation of adequate size and quality of training samples. Neural networks are good at operating with implicit data, generalizing through learning process but approximation results on feed-forward networks depend drastically [4, 7] on the quality of training sample. In real world comprehensive data in most cases is unavailable.

Briefly speaking classical NN represent parallel implementation of corresponding pattern recognition methods by algebraic or probabilistic computations and thus inherit their advantages and most of disadvantages.

What can be used as integrator of different levels of knowledge about very complex system about brain? One of possible answers “synergy” science about systems self-organization [6, 18]. And use of synergy for development of AI theory is rather natural if we take into account that brain is dynamic system. Because of this there is growth of interest to bio-inspired and dynamic approaches today.

3 Novel Bio-inspired Methods and Dynamic Neural Networks

In spite of the fact that idea of artificial intelligence is already bio-inspired (like in the nature) the approximation to biological prototype can differ in the process of AI development. Classical NN models are comprised of formal neurons. Now more and more attention is given to bio-inspired NN with neurons similar to nerve cells. Recent investigations speak for the sputter out phase

in NN models design as the solutions quality improvements are insufficient in comparison to huge efforts spent in this direction. Most likely detailed reproduction of the processes occurring in separate nerve cells without understanding the principals of mutual interactions will result with analogous effect.

Obtained results supplement our knowledge on multiform behavior of biologic systems but simple reproduction of neuron ensembles characteristics for partial regimes is hardly to be generalized. It is underlined that improvement of one part of parameters negatively influence on other previously well-tuned. Bottom-up de-sign starting from the lowest level most likely will not result with emergence of system with new qualities. Synergy effects occur mainly when self-organization principals underlie the system [6, 21].

There are two classes of NN: static (nearly all possible results here are reached already) and dynamic (interest to them is growing today). The evolution curve of dynamic NN can be separated by stages:

1. First stage – appearing of Hopfield’s and then Haken’s NN – attractors NN. Attractors in these dynamic NN are simple enough – fix-point attractors [6, 7].
2. Second stage – dynamic NN with more complex cycle attractors [22].
3. Current stage – dynamic NN with unstable dynamics – most complex chaotic attractors. Their dynamic is characterized by set of trajectories scaled in the phase space to a location with infinite number of switching states [2, 13].

Separate subclass of current stage dynamic NN are NN of reservoir computing [12, 13]. The origins of such systems can be found starting from second stage - they had random structure capable to generate promising complex dynamics.

When speaking about reservoir computing one of the main difficulties is the absence of guarantee that system will produce the required dynamics because there exists a great uncertainty about reservoir parameters (especially actual for Maass bio-inspired model). Empiric (or intuitive) parameter assignment is done on the basis of preliminary experiments series. There is strong proof that Turing machine can be realized on reservoir NN [13] but wide practical application of this approach is still the matter of future investigations.

To develop network with complex dynamics basic transfer functions in recurrent NN are replaced with chaotic maps [2, 16]. Application of functions that provide deterministic chaos corresponds with the current trend to combine linkage complexity concept with gradual complication of processing units. The most applicable is one-dimension logistic map that allows to control degree of generated chaos by means of one parameter.

The development of unified approach to AI requires to examine not isolated system but system in the environment where this system will be used. This idea is widely discussed in [17] where role of chaos intelligent agent development is considered. Most likely that such holistic approach will help to formalize at last such complex notion as context.

Just chaotic systems, whose dynamic strongly depends from external conditions, can help to represent context and as well all variety of possible situations (of really mobile and often preliminary undefined environment) in which tasks should be solved. Some promising results are obtained already [2, 6, 13, 17, 18, 21, 22].

4 Conclusions

Development of intellectual systems specified to solve certain class of problems should be obviously accomplished by means of approved methods or their hybrid modifications. Detailed reproduction of separate neuron cell ensembles in attempt to investigate some cortex domain is rational in neurophysiology models but not in AI most probably. Unified approach to development of artificial intelligent systems with quality commensurable to natural neural systems to our opinion should be based on distributed ensembles of chaotic (coupled) maps. This direction of research is attractive as it combines ideas of agent theory, NN theory, nonlinear dynamics, synchronization theory, formal logics. Consequently ample opportunities to apply complex synergetic effects to deal with uncertainty not only in technical but also in biological, economic, geopolitical systems foster an idea that in the nearest future nonlinear dynamics and chaos will become the most demanded apparatus to understand and model cognition processes.

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